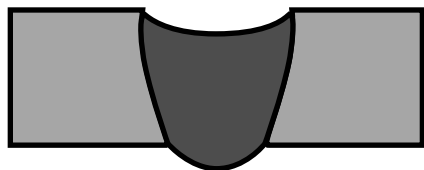


Stick Weld Penetration (Workpiece Heat Control)

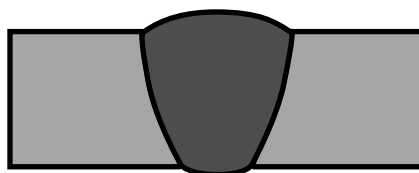
EXCESS PENETRATION OR BURN-THROUGH

Weld droops on top and underneath or falls through entirely, making a hole.



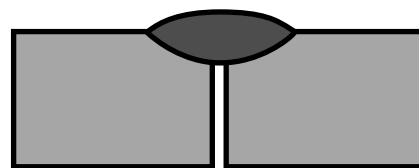
PROPER PENETRATION

Weld is visible underneath and bulges slightly on top.



INADEQUATE PENETRATION

Weld does not contact the joint fully, just on the surface.



PROFILE VIEWS

POSSIBLE CAUSES AND SOLUTIONS FOR EXCESS PENETRATION OR BURN-THROUGH

1. **Workpiece overheating:**
Reduce current.
2. **Welding speed too slow:**
Increase welding speed and ensure that welding speed is kept steady.

POSSIBLE CAUSES AND SOLUTIONS FOR INADEQUATE PENETRATION

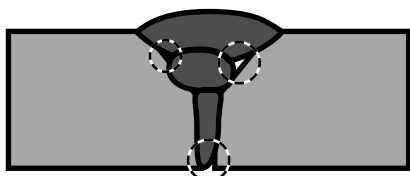
1. **Incorrect welding technique:**
Keep arc on leading edge of weld puddle. Hold torch at proper angles.
2. **Insufficient weld heat:**
Slow down so fill material has time to melt into the weld location. Increase current.
3. **Workpieces too thick/close:**
Bevel thick workpieces, allow slight gap, and weld in several passes.
4. **Insufficient weld material:**
Increase amount of fill material.

Stick Weld–Weld Not Adhering Properly

Gaps present between weld and previous bead or between weld and workpiece. See areas below.

POSSIBLE CAUSES AND SOLUTIONS

PROFILE VIEW



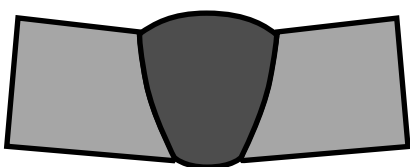
1. **Incorrect welding technique:**
Place stringer bead at correct place in joint. Adjust workpiece position or weld angle to permit proper welding to bottom of piece. Keep arc on leading edge of weld puddle. Hold Electrode and fill material at proper angles.

2. **Insufficient weld heat:**
Increase current.
3. **Dirty workpiece:**
Clean workpiece down to bare metal.
4. **Insufficient weld material:**
Increase amount of fill material.
5. **Distance between workpieces too large:**
Decrease distance and increase bevel.

Stick Weld–Bend at Joint

POSSIBLE CAUSES AND SOLUTIONS

PROFILE VIEW



1. **Improper clamping:**
Clamp workpieces securely. Make tack welds to hold workpieces.
2. **Excessive heat:**
Weld a small portion and allow to cool before proceeding. Increase weld speed.